

chips passing through a global supply chain, interception and tampering can occur, adding another layer of risk.

Why are chip backdoors so hard to detect?

Detecting these vulnerabilities is a formidable challenge for several reasons:

Invisible to software. Traditional security tools, like antivirus software and firewalls, are powerless against hardware-based backdoors. They operate outside the digital boundaries that software usually monitors.

Complexity of chip design. Modern chips are incredibly complex, featuring billions of transistors and circuits. This intricate structure makes thorough inspection impractical and leaves ample room for vulnerabilities to hide.

Conditional activation. Many backdoors are designed to activate under specific and often obscure conditions. This means they can remain dormant for long periods, sidestepping regular testing and usage detection.

Notable cases of chip backdoors

While most cases remain classified, some high-profile instances have spotlighted the issue:

Bloomberg's 'The Big Hack' Report. In 2018, Bloomberg Businessweek reported that Chinese spies had allegedly implanted microchips into servers used by major US companies such as Amazon and Apple. Although both companies strongly denied the allegations, the report sparked a global conversation about supply chain vulnerabilities.

NSA hardware backdoors. Leaks from Edward Snowden showed that the US National Security Agency (NSA) had partnered with manufacturers to insert hardware backdoors, enabling surveillance of foreign targets.

Military equipment concerns. Claims of compromised chips in military hardware have raised alarms about the security of national defence systems and the risk of cyber espionage.

Counteracting the threat: Steps we can take

Preventing chip backdoors requires a multifaceted approach. Here are some of the key measures:

Enhancing supply chain security. Collaborating with trusted suppliers and strengthening oversight are essential steps for reducing risks. Governments and private firms need to work together to build a more secure pipeline for chip production.

Comprehensive chip audits. Rigorous hardware checks, including specialised audits and physical inspections, should be part of standard practices. Independent verification can further ensure chip integrity.

Design redundancy. Employing a strategy where multiple suppliers produce similar chips helps minimise the impact of a compromised batch.

Setting global standards. International cooperation on hardware security standards can play a significant role in safeguarding chip production processes.

Post-manufacturing safeguards. Using advanced encryption and designing chips to be tamper-resistant can add layers of security, protecting critical operations within the hardware.

The insecurity wave

Spectre and Meltdown are vulnerabilities that highlight the fine line between chip performance and security. These exploits leverage speculative execution, a feature enabling processors to anticipate and execute instructions ahead of time, to gain access to sensitive data. Spectre targets isolation between applications, while Meltdown breaks the security boundary between user programs and the system kernel.

Through well-crafted code, attackers can extract confidential data by observing timing discrepancies in the processor's operations. While patches exist, fully addressing these vulnerabilities without affecting

performance has proven challenging, showcasing the delicate balance we must maintain.

Trojan code embedded at the chip's design level can create backdoors that bypass conventional security defences. This type of Trojan can be introduced before manufacturing, embedding malicious circuits or altering logic to enable data extraction or device control. As these changes occur at the hardware level, they are extremely difficult to detect.

The growing complexity of chip and chiplet design exacerbates the challenge. Integrating multiple components and third-party intellectual property (IP) blocks increases the risk of unintentional or deliberate vulnerabilities being inserted. Verifying trust at every stage becomes more difficult as both manual reviews and automated tools struggle with intricate chip architectures.

Why we should be concerned

The threat of chip backdoors extends beyond personal privacy to encompass national security. A single compromised chip can provide access to sensitive data across personal, corporate, and government sectors. As our dependence on digital systems deepens, ensuring trust in these devices is critical for maintaining economic stability and strategic advantage.

The intricate nature of modern chips means complete assurance against tampering is challenging. Any breach in trust could hinder innovation, strain international relations, and elevate geopolitical tension.

The role of chips in our world is only expanding, and with it comes the risk of chip backdoors. Tackling these threats requires enhanced security practices, careful oversight, and international collaboration. A proactive approach, coupled with awareness, is our first line of defence against this hidden but potent danger to our digital infrastructure. **EFY**